Lecture 03: Exploring and visualising data

ENVX1002 Statistics in Life and Environmental Sciences

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Learning outcomes

After this week, you will be able to:

1. Explore datasets using summary functions and identify data quality issues
2. Choose appropriate plot types for different kinds of data
3. Build layered visualisations in R using ggplot2
4. Interpret distribution shapes, including skewness and kurtosis

Quick checklist

By now you should have...

- ☐ Installed **R** and **RStudio**
- ☐ Completed Lecture 2 content and read the ENVX1002 R guide
- ☐ Reviewed measures of central tendency and spread
- ☐ Practised using R functions with the `function(argument = value)` syntax
- ☐ Rendered a few Quarto documents in RStudio

Why explore data first?

Looking before you leap

Jumping straight into analysis without exploring your data is like navigating without a map. You might arrive somewhere, but probably not where you intended.

Data exploration helps you:

- Spot problems early (missing values, outliers, errors)
- Understand what shape your data takes
- Choose the right analysis method
- Avoid drawing conclusions from flawed data

The data exploration workflow

A practical sequence for getting to know a new dataset:

1. Check the structure and data types
2. Summarise with descriptive statistics
3. Visualise distributions and relationships
4. Flag anything unexpected

Quick recap: types of data

R needs to know what kind of data it is working with. Here is the breakdown:

	Type	Order?	Example
Categorical	Nominal	No	Species, blood type
	Ordinal	Yes	Pain scale (mild ⟶ severe)
Continuous	Interval	—	Temperature in °C
	Ratio	—	Height, weight, age

Categorical data groups observations into labels. Continuous data measures quantities on a scale. The distinction matters because each type calls for different summaries and different plots.

How is your data shaped?

> "The greatest value of a picture is when it forces us to notice what we never expected to see." — John Tukey, *Exploratory Data Analysis* (1977)

The shape of your data influences everything: which summary statistics make sense, which plots to use, and which analyses are valid.

The normal distribution

The most common distribution you will encounter is the normal (or "bell curve") distribution. It is symmetric, with most values clustered near the centre and fewer in the tails.

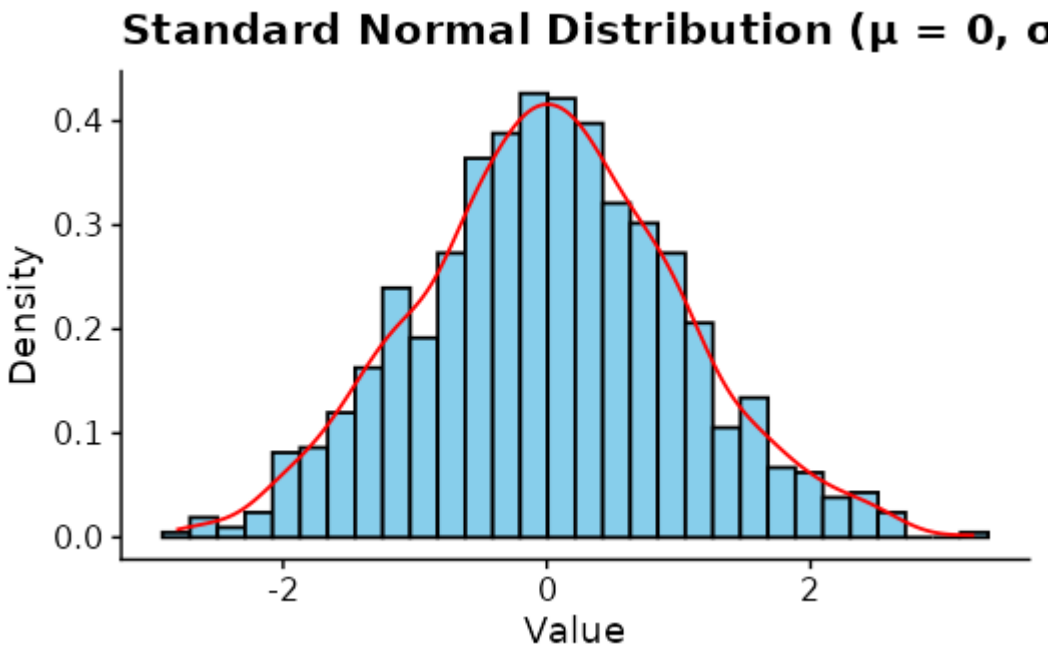
- Defined by two parameters: the mean (μ) and standard deviation (σ)
- Appears frequently in nature: heights, measurement errors, physiological traits

$$X \sim N(\mu, \sigma^2)$$

This notation means: the variable X follows a normal distribution with mean μ and variance σ^2.

What does a normal distribution look like?

```r
# Generate normal distribution data
set.seed(123)
normal_data <- rnorm(1000, mean = 0, sd = 1)

# Plot normal distribution
ggplot(data.frame(x = normal_data), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30,
                 fill = "skyblue",
                 colour = "black") +
  geom_density(colour = "red") +
  labs(title = "Standard Normal Distribution (μ = 0, σ = 1)",
       x = "Value",
       y = "Density")
```


Standard Normal Distribution (μ = 0, σ
Density
0.0
0.1
0.2
0.3
0.4
-2
0
2
Value

The empirical rule (68-95-99.7)

For a normal distribution, the mean, median, and mode are all the same value. The data clusters predictably around that centre:

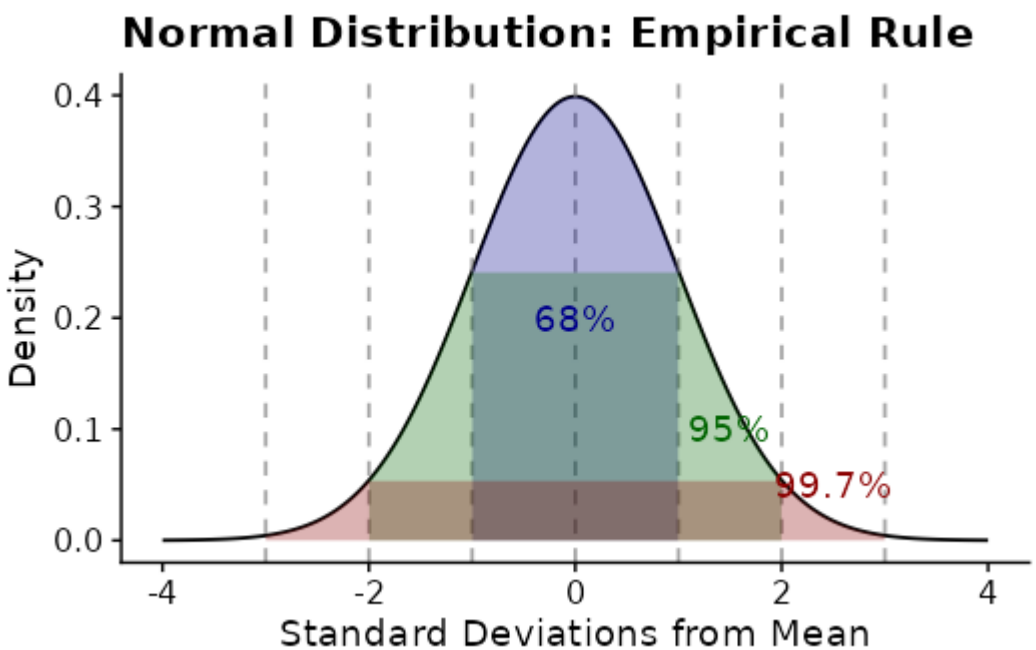
- About **68%** of values fall within 1 standard deviation of the mean
- About **95%** within 2 standard deviations
- About **99.7%** within 3 standard deviations

The empirical rule, visualised

This pattern makes the normal distribution especially useful: if you know the mean and standard deviation, you already know where most of the data sits.

```
# Create a standard normal distribution
x <- seq(-4, 4, length.out = 1000)
y <- dnorm(x)
df <- data.frame(x = x, y = y)

# Plot with empirical rule highlighted
ggplot(df, aes(x = x, y = y)) +
  geom_line() +
  # Add vertical reference lines at standard deviations
  geom_vline(xintercept = c(-3, -2, -1, 0, 1, 2, 3), linetype = "dashed", colour = "gray50", alpha
= 0.7) +
  geom_area(data = subset(df, x ≥ -1 & x ≤ 1), fill = "darkblue", alpha = 0.3) +
  geom_area(data = subset(df, (x ≥ -2 & x < -1) | (x > 1 & x ≤ 2)), fill = "darkgreen", alpha =
0.3) +
  geom_area(data = subset(df, (x ≥ -3 & x < -2) | (x > 2 & x ≤ 3)), fill = "darkred", alpha =
0.3) +
```

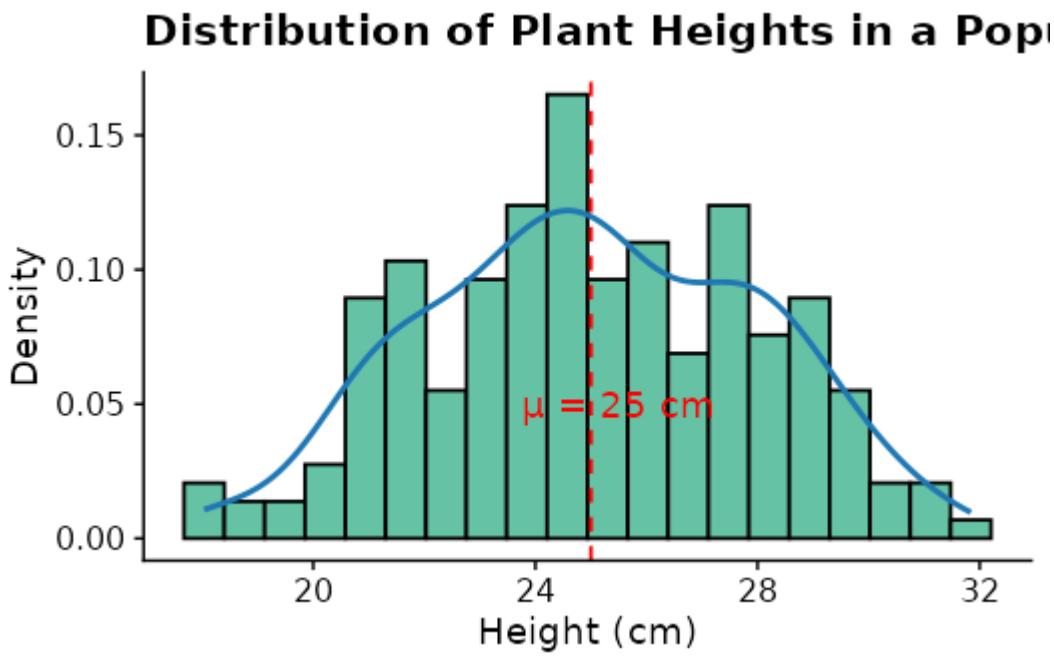
```r
  annotate("text", x = 0, y = 0.2, label = "68%", colour = "darkblue") +
  annotate("text", x = 1.5, y = 0.1, label = "95%", colour = "darkgreen") +
  annotate("text", x = 2.5, y = 0.05, label = "99.7%", colour = "darkred") +
  labs(title = "Normal Distribution: Empirical Rule",
       x = "Standard Deviations from Mean",
       y = "Density")
```



Why does this matter?

Many biological measurements follow a normal distribution. If you know your data is roughly normal, you can predict where most values will land and spot anything unusual.

```r
# Simulate plant height data
set.seed(456)
plant_heights <- rnorm(200, mean = 25, sd = 3)  # Heights in cm

# Plot the distribution
ggplot(data.frame(height = plant_heights), aes(x = height)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 20,
                 fill = "#66c2a5",
                 colour = "black") +
  geom_density(colour = "#1f78b4", linewidth = 1) +
  geom_vline(xintercept = 25, linetype = "dashed", colour = "red") +
  annotate("text", x = 25.5, y = 0.05, label = "μ = 25 cm", colour = "red") +
  labs(title = "Distribution of Plant Heights in a Population",
       x = "Height (cm)",
       y = "Density")
```



Here, plant heights cluster around a mean of 25 cm. A plant measuring 35 cm would be more than 3 standard deviations from the mean, making it a clear outlier worth investigating.

But what if the data is not symmetric?

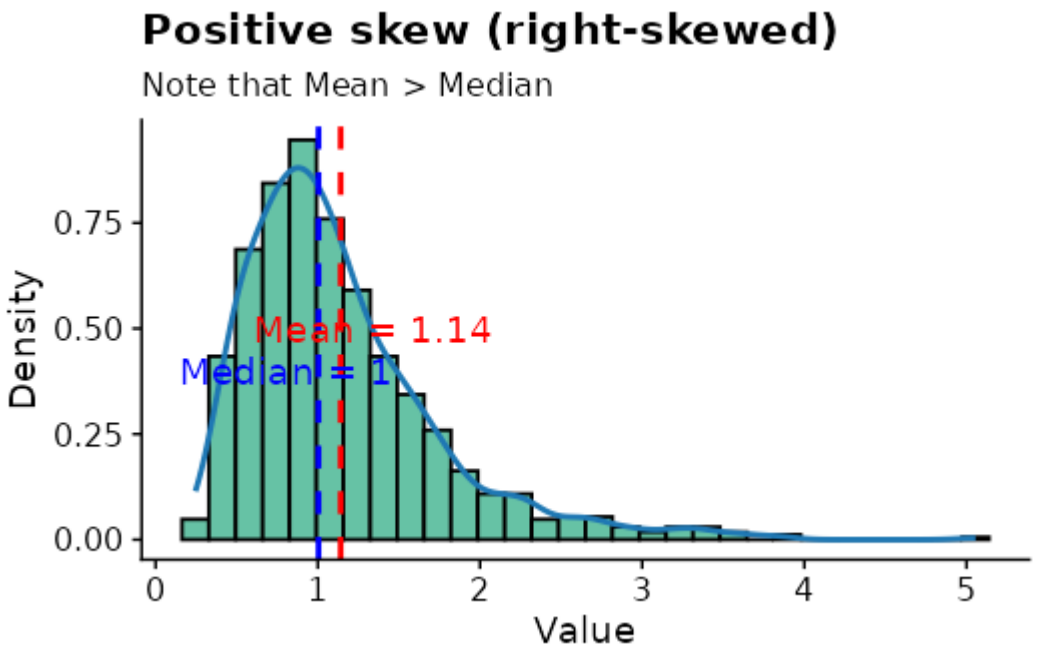
Not all data follows a neat bell curve. When a distribution leans to one side, we call it **skewed**. The direction of the longer tail tells you the type of skew, and it shifts the mean away from the median.

Positive skew (right-skewed)

The tail stretches to the right, pulling the mean above the median. Common in income data, reaction times, and species abundance counts.

```
# Generate positive skewed distribution
set.seed(123)
right_skewed <- exp(rnorm(1000, 0, 0.5))

# Calculate statistics
mean_val <- mean(right_skewed)
median_val <- median(right_skewed)

# Plot positive skewed distribution
ggplot(data.frame(x = right_skewed), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30,
                 fill = "#66c2a5", # colourblind-friendly green
                 colour = "black") +
  geom_density(colour = "#1f78b4", linewidth = 1) +
  # Add vertical lines for mean and median
```

```r
  geom_vline(xintercept = mean_val, colour = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = median_val, colour = "blue", linetype = "dashed", linewidth = 1) +
  # Add annotations
  annotate("text", x = mean_val + 0.2, y = 0.5, label = paste("Mean =", round(mean_val, 2)), colour
= "red") +
  annotate("text", x = median_val - 0.2, y = 0.4, label = paste("Median =", round(median_val, 2)),
colour = "blue") +
  labs(title = "Positive skew (right-skewed)",
       subtitle = "Note that Mean > Median",
       x = "Value",
       y = "Density")
```

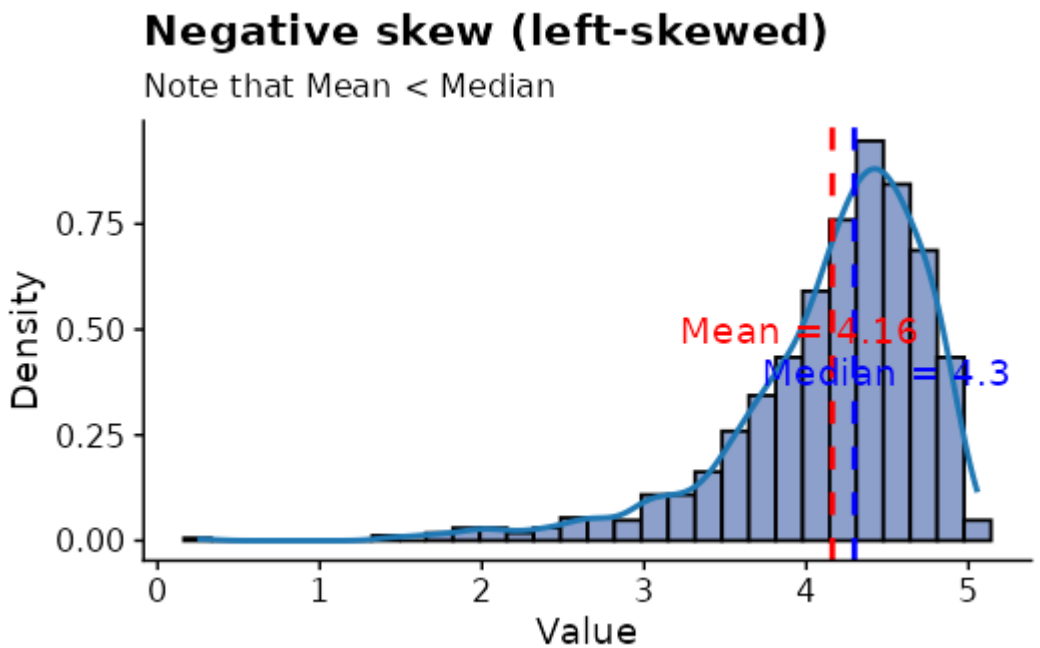

Positive skew (right-skewed)
Note that Mean > Median
Mean = 1.14
Median = 1
Density
0.00
0.25
0.50
0.75
0
1
2
3
4
5
Value

Negative skew (left-skewed)

The tail stretches to the left, pulling the mean below the median. Less common, but seen in exam scores near a ceiling or age at retirement.

```
# Generate skewed distributions
set.seed(123)
left_skewed <- max(right_skewed) - right_skewed + min(right_skewed)

# Calculate statistics
mean_val <- mean(left_skewed)
median_val <- median(left_skewed)

# Plot negative skewed distribution
ggplot(data.frame(x = left_skewed), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)),
                 bins = 30,
                 fill = "#8da0cb", # colourblind-friendly blue
                 colour = "black") +
  geom_density(colour = "#1f78b4", linewidth = 1) +
  # Add vertical lines for mean and median
```

```r
  geom_vline(xintercept = mean_val, colour = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = median_val, colour = "blue", linetype = "dashed", linewidth = 1) +
  # Add annotations
  annotate("text", x = mean_val - 0.2, y = 0.5, label = paste("Mean =", round(mean_val, 2)), colour
= "red") +
  annotate("text", x = median_val + 0.2, y = 0.4, label = paste("Median =", round(median_val, 2)),
colour = "blue") +
  labs(title = "Negative skew (left-skewed)",
       subtitle = "Note that Mean < Median",
       x = "Value",
       y = "Density")
```


Negative skew (left-skewed)
Note that Mean < Median
Mean = 4.16
Median = 4.3
Density
0.00
0.25
0.50
0.75
0
1
2
3
4
5
Value

Kurtosis: how heavy are the tails?

Kurtosis describes whether a distribution has heavy tails (more extreme values than expected) or light tails (fewer extremes). It is not about how "peaked" the distribution looks, despite what some textbooks say.

- **Mesokurtic**: a normal distribution (the baseline)
- **Leptokurtic**: heavier tails, more outliers
- **Platykurtic**: lighter tails, fewer outliers

Kurtosis, visualised

Compare these three distributions. They all have the same mean and similar spread, but their tails behave very differently.

```r
# Generate distributions with different kurtosis
set.seed(123)
normal <- rnorm(1000, 0, 1)  # Mesokurtic
leptokurtic <- rt(1000, df = 5)  # t-distribution with 5 df is leptokurtic
platykurtic <- runif(1000, -3, 3)  # Uniform distribution is platykurtic

# Calculate kurtosis values (using e1071 package)
library(e1071)
```

```
Attaching package: 'e1071'
```

```
The following object is masked from 'package:ggplot2':

    element
```

```r
k_normal <- kurtosis(normal)
k_lepto <- kurtosis(leptokurtic)
k_platy <- kurtosis(platykurtic)

# Plot distributions
p1 <- ggplot(data.frame(x = normal), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "#a6cee3", # colourblind-friendly blue
                 colour = "black") +
  geom_density(colour = "#1f78b4", linewidth = 1) + # Darker blue
  annotate("text", x = 2, y = 0.3,
           label = paste("Kurtosis =", round(k_normal, 2)),
           colour = "#1f78b4") +
  labs(title = "Mesokurtic (normal)",
       subtitle = "Normal distribution with balanced tails",
       x = "Value",
       y = "Density")

p2 <- ggplot(data.frame(x = leptokurtic), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "#fb9a99", # colourblind-friendly pink
```

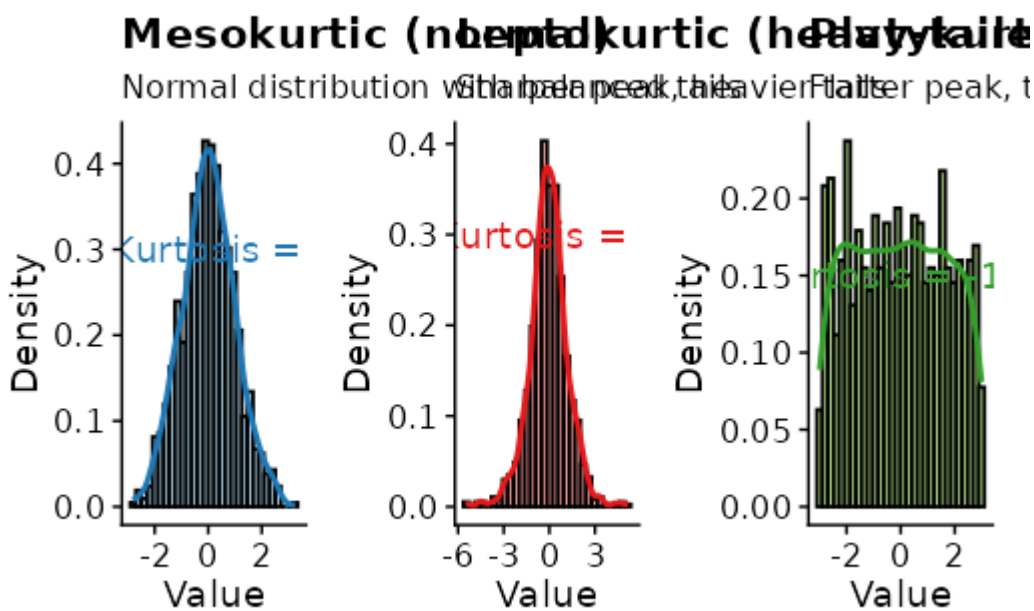
```r
                 colour = "black") +
  geom_density(colour = "#e31a1c", linewidth = 1) + # Darker red
  annotate("text", x = 2, y = 0.3,
           label = paste("Kurtosis =", round(k_lepto, 2)),
           colour = "#e31a1c") +
  labs(title = "Leptokurtic (heavy-tailed)",
       subtitle = "Sharper peak, heavier tails",
       x = "Value",
       y = "Density")

p3 <- ggplot(data.frame(x = platykurtic), aes(x = x)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30,
                 fill = "#b2df8a", # colourblind-friendly green
                 colour = "black") +
  geom_density(colour = "#33a02c", linewidth = 1) + # Darker green
  annotate("text", x = 0, y = 0.15,
           label = paste("Kurtosis =", round(k_platy, 2)),
           colour = "#33a02c") +
  labs(title = "Platykurtic (light-tailed)",
       subtitle = "Flatter peak, thinner tails",
       x = "Value",
```

```r
       y = "Density")

# Display plots vertically
library(patchwork)
```

```
Attaching package: 'patchwork'
```

```
The following object is masked from 'package:cowplot':

    align_plots
```

```r
p1 + p2 + p3
```


Mesokurtic (n
Normal distribution w
Density
Value
Kurtosis =
Density
Value
urtosis =
Density
Value
Kurtosis = -1

Exploring data in R

Data structures you will use

The two structures you will work with most in this course:

Vectors: a single row of values

```
heights <- c(1.65, 1.70, 1.75, 1.80, 1.85)
heights
```

```
[1] 1.65 1.70 1.75 1.80 1.85
```

Data frames: a table with rows and columns (like a spreadsheet)

```
df <- data.frame(
  species = c("A", "B", "C", "A", "B"),
  height = c(1.65, 1.70, 1.75, 1.80, 1.85),
  weight = c(60, 65, 70, 75, 80)
)
df
```

```
  species height weight
1       A   1.65     60
2       B   1.70     65
3       C   1.75     70
4       A   1.80     75
5       B   1.85     80
```

Getting to know your data

A few functions that give you an immediate overview:

```r
str(df)
```

```
'data.frame':	5 obs. of  3 variables:
 $ species: chr  "A" "B" "C" "A" ...
 $ height : num  1.65 1.7 1.75 1.8 1.85
 $ weight : num  60 65 70 75 80
```

```r
summary(df)
```

```
   species              height         weight
 Length:5           Min.   :1.65   Min.   :60
 Class :character   1st Qu.:1.70   1st Qu.:65
 Mode  :character   Median :1.75   Median :70
                    Mean   :1.75   Mean   :70
                    3rd Qu.:1.80   3rd Qu.:75
                    Max.   :1.85   Max.   :80
```

```r
unique(df$species)
```

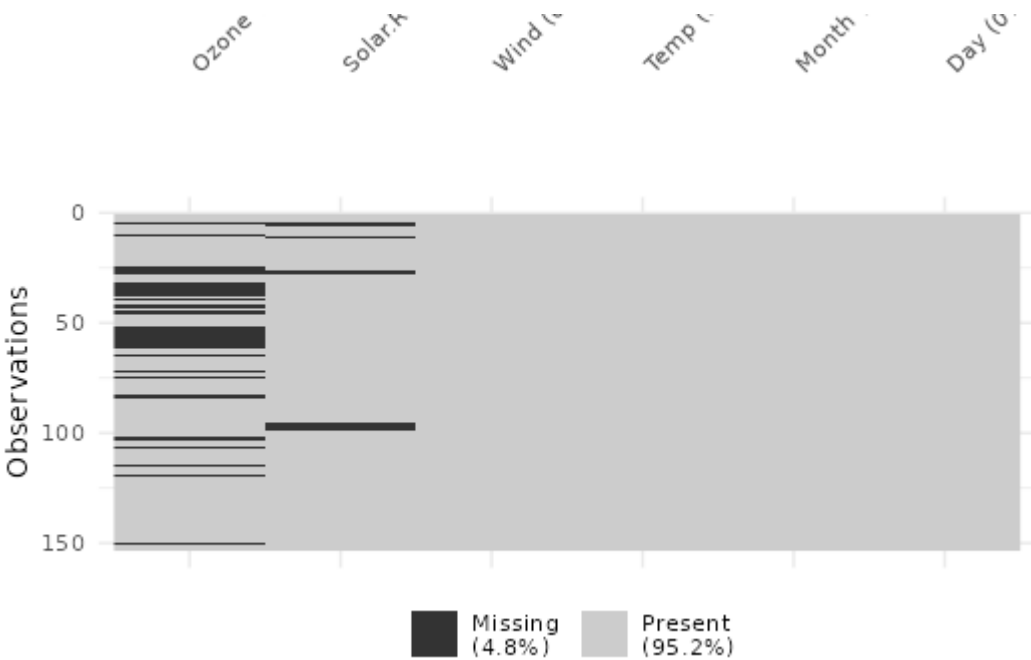
```
[1] "A" "B" "C"
```

`summary()` is useful, but it does not always reveal the full picture.

Visualising missing data

The `naniar` package can show you exactly where values are missing, which is much easier to interpret than scanning raw output.

```
library(naniar)
vis_miss(airquality)
```


Ozone
Solar.R
Wind
Temp
Month
Day
Observations
0
50
100
150
Missing
(4.8%)
Present
(95.2%)

Seeing vs reading

Compare the `vis_miss()` output to the raw data below. Which one makes the missing values easier to spot?

```
airquality$Ozone
```

```
  [1]  41  36  12  18  NA  28  23  19   8  NA   7  16  11  14  18  14  34   6
 [19]  30  11   1  11   4  32  NA  NA  NA  23  45 115  37  NA  NA  NA  NA  NA
 [37]  NA  29  NA  71  39  NA  NA  23  NA  NA  21  37  20  12  13  NA  NA  NA
 [55]  NA  NA  NA  NA  NA  NA  NA 135  49  32  NA  64  40  77  97  97  85  NA
 [73]  10  27  NA   7  48  35  61  79  63  16  NA  NA  80 108  20  52  82  50
 [91]  64  59  39   9  16  78  35  66 122  89 110  NA  NA  44  28  65  NA  22
[109]  59  23  31  44  21   9  NA  45 168  73  NA  76 118  84  85  96  78  73
[127]  91  47  32  20  23  21  24  44  21  28   9  13  46  18  13  24  16  13
[145]  23  36   7  14  30  NA  14  18  20
```

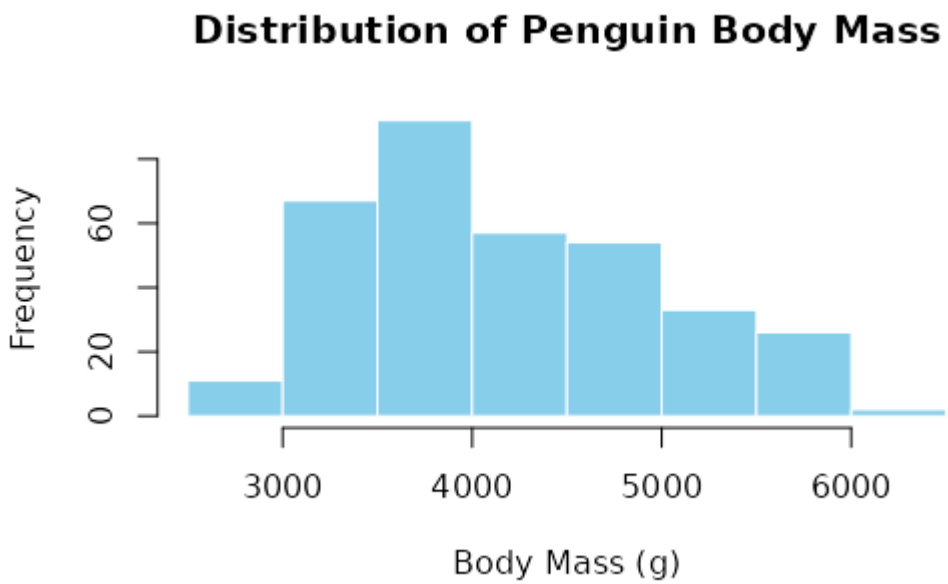
```
summary(airquality$Ozone)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's 
   1.00   18.00   31.50   42.13   63.25  168.00      37 
```

Histograms: the shape of continuous data

Use histograms when you want to see how a continuous variable is distributed. They reveal the centre, spread, and any skewness or outliers at a glance.

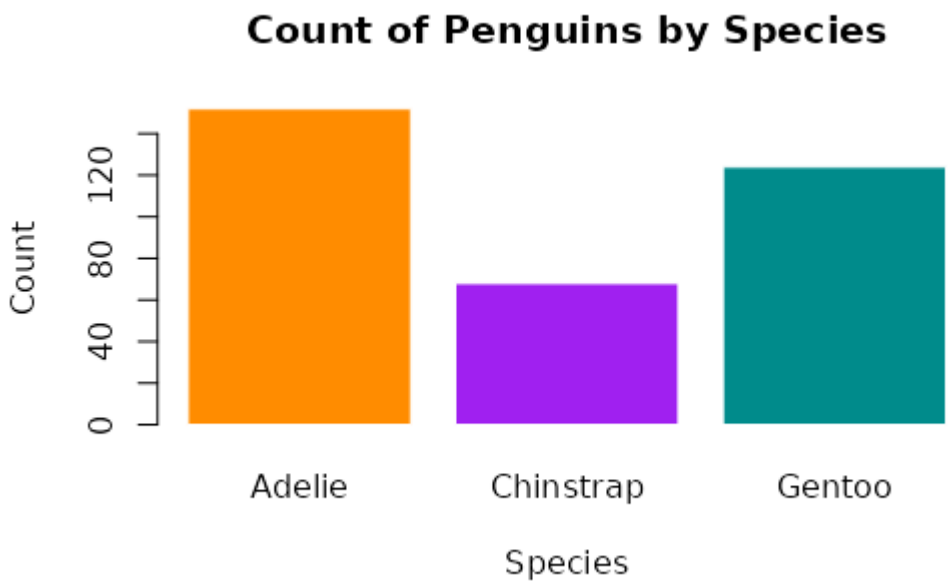
```r
hist(penguins$body_mass_g,
     main = "Distribution of Penguin Body Mass",
     xlab = "Body Mass (g)",
     col = "skyblue",
     border = "white")
```


Distribution of Penguin Body Mass
Frequency
0
20
60
3000
4000
5000
6000
Body Mass (g)

Bar plots: comparing categories

Bar plots show counts or proportions across groups. Use them for categorical data like species, treatments, or survey responses.

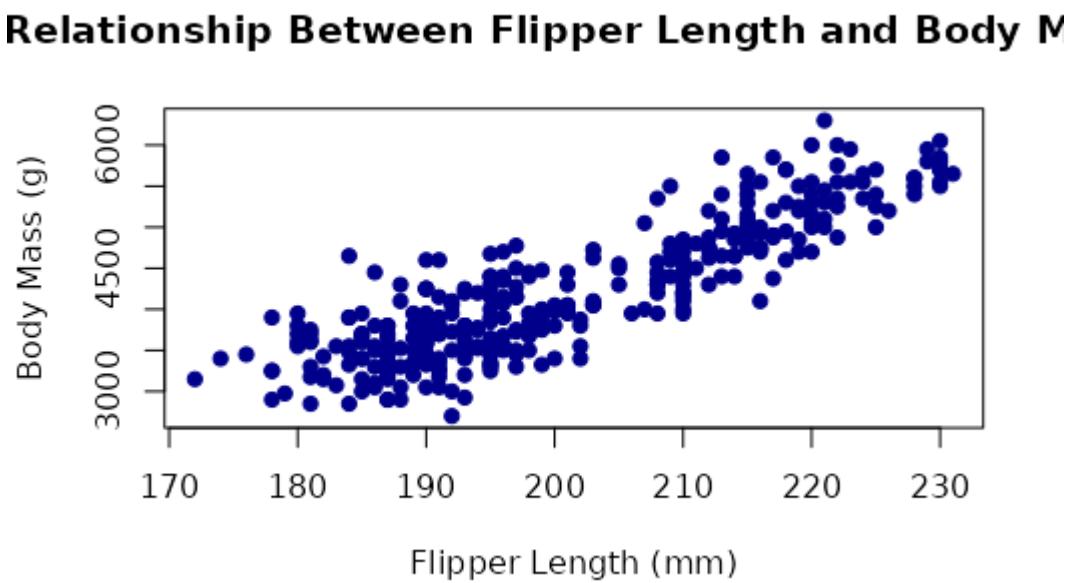
```
species_counts <- table(penguins$species)
barplot(species_counts,
        main = "Count of Penguins by Species",
        xlab = "Species", ylab = "Count",
        col = c("darkorange", "purple", "cyan4"),
        border = "white")
```


Count of Penguins by Species
Count
0
40
80
120
Adelie
Chinstrap
Gentoo
Species

Scatterplots: relationships between variables

Scatterplots show how two continuous variables relate to each other. They are the go-to plot for spotting correlations, clusters, and outliers.

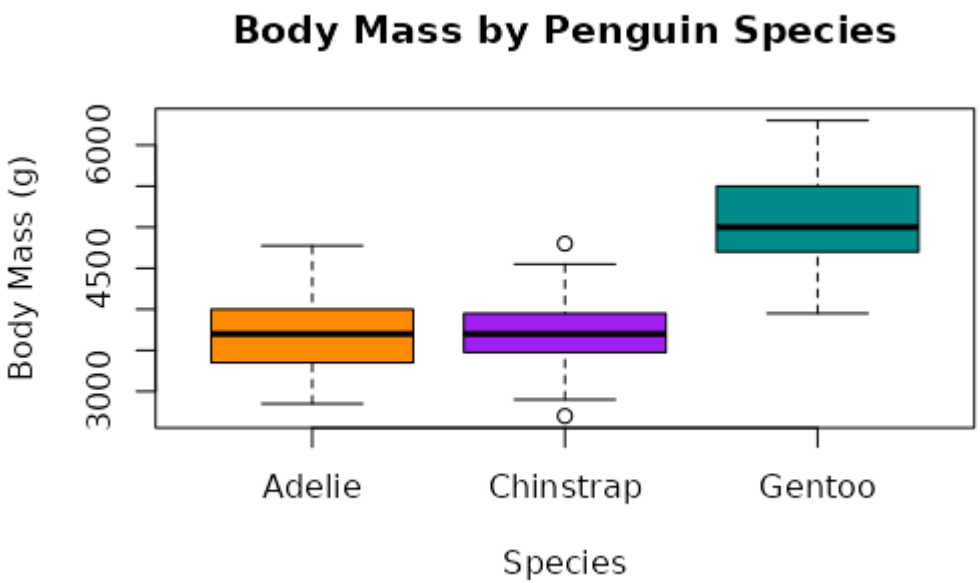
```
penguins_clean <- na.omit(penguins[, c("flipper_length_mm", "body_mass_g")])
plot(penguins_clean$flipper_length_mm, penguins_clean$body_mass_g,
     main = "Relationship Between Flipper Length and Body Mass",
     xlab = "Flipper Length (mm)", ylab = "Body Mass (g)",
     pch = 19, col = "darkblue")
```


Relationship Between Flipper Length and Body M
Body Mass (g)
3000
4500
6000
170
180
190
200
210
220
230
Flipper Length (mm)

Boxplots: comparing distributions across groups

Boxplots summarise the median, quartiles, and outliers for each group. They are ideal for comparing a continuous variable across categories.

```
boxplot(body_mass_g ~ species, data = penguins,
        main = "Body Mass by Penguin Species",
        xlab = "Species", ylab = "Body Mass (g)",
        col = c("darkorange", "purple", "cyan4"),
        border = "black")
```


Body Mass by Penguin Species
Body Mass (g)
3000
4500
6000
Adelie
Chinstrap
Gentoo
Species

Introduction to ggplot2

A grammar for graphics

ggplot2 builds plots from three core ingredients:

1. **Data**: the dataset
2. **Aesthetics** (`aes`): which variables map to x, y, colour, size, etc.
3. **Geometries** (`geom_*`): how the data is drawn (points, bars, lines, etc.)

You combine these with `+` to build a plot layer by layer. As you get more comfortable, you can add scales, facets, and themes to refine the result.

Building a plot: start with data

We will build a scatterplot of penguin flipper length vs body mass, step by step.

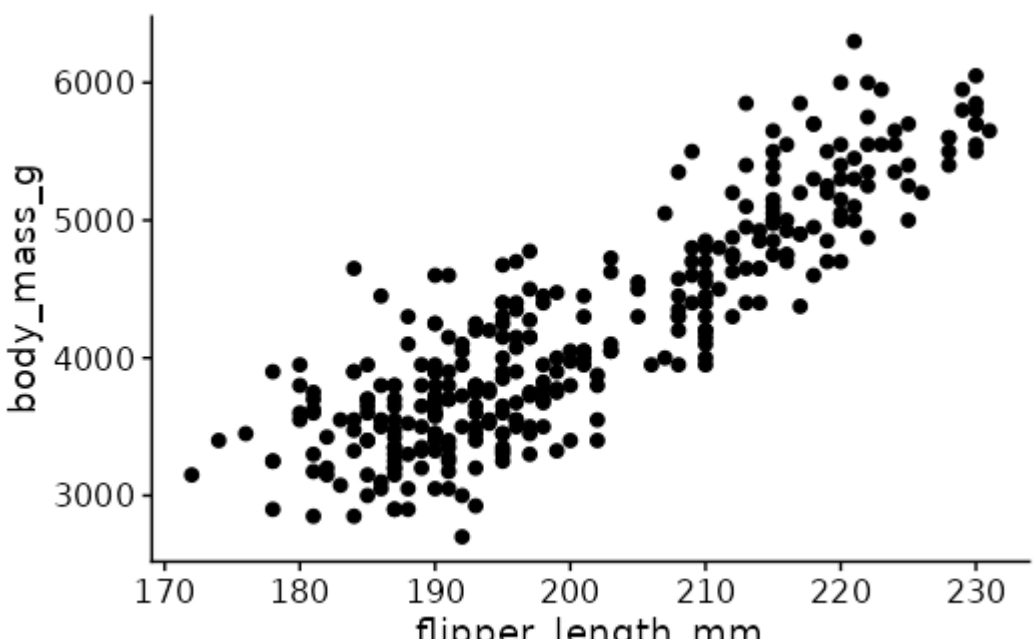
```
library(ggplot2)
penguins_clean <- na.omit(penguins)

# Start with data — this creates an empty canvas
ggplot(penguins_clean)
```

Nothing to see yet. We need to tell ggplot2 *what* to plot and *how* to draw it.

Add aesthetics and a geometry

Map variables to axes with `aes()`, then choose a geometry to draw them:

```
ggplot(penguins_clean, aes(x = flipper_length_mm, y = body_mass_g)) +
  geom_point()
```

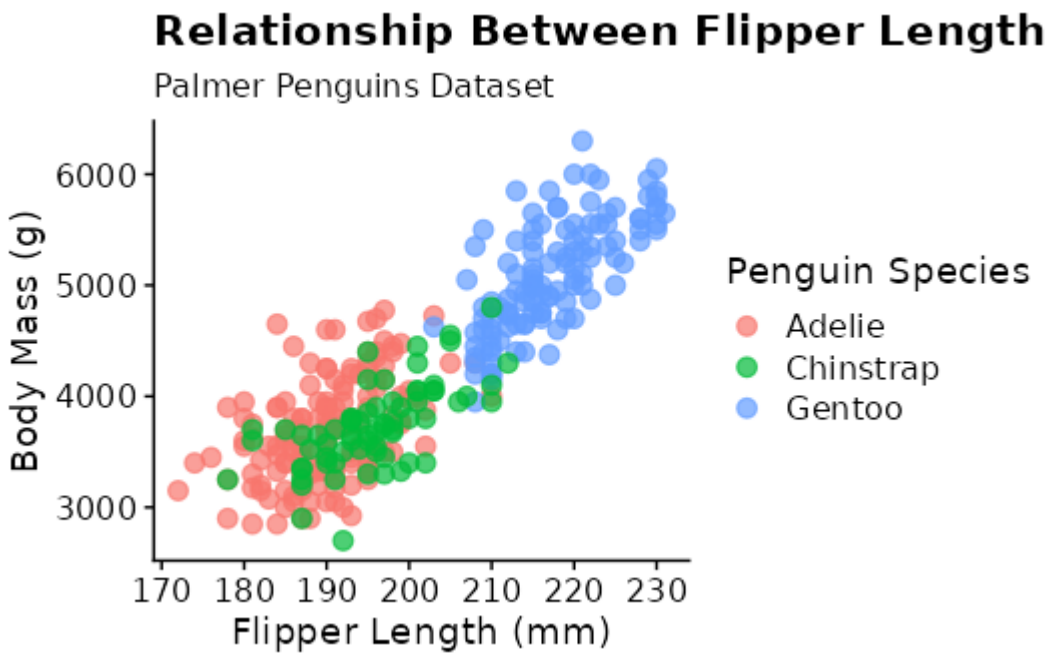


Two lines of code, and we already have a working scatterplot. The `+` operator is how you add layers in ggplot2.

Colour by group and add labels

Map a categorical variable to `colour` inside `aes()` to split by group. Use `labs()` to add titles and clean up axis names.

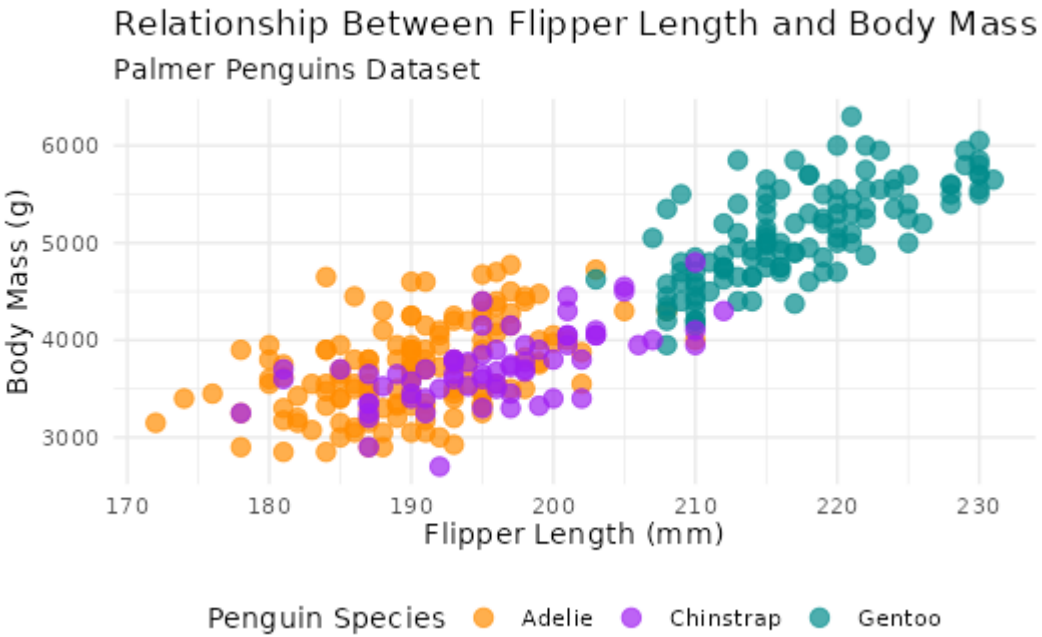
```
ggplot(penguins_clean, aes(x = flipper_length_mm, y = body_mass_g, colour = species)) +
  geom_point(size = 3, alpha = 0.7) +
  labs(
    title = "Relationship Between Flipper Length and Body Mass",
    subtitle = "Palmer Penguins Dataset",
    x = "Flipper Length (mm)",
    y = "Body Mass (g)",
    colour = "Penguin Species"
  )
```


Relationship Between Flipper Length
Palmer Penguins Dataset
Body Mass (g)
6000
5000
4000
3000
170
180
190
200
210
220
230
Flipper Length (mm)
Penguin Species
Adelie
Chinstrap
Gentoo

Customise colours and themes

`scale_colour_manual()` lets you pick your own palette. Themes control the overall look of the plot.

```
p <- ggplot(penguins_clean, aes(x = flipper_length_mm, y = body_mass_g, colour = species)) +
  geom_point(size = 3, alpha = 0.7) +
  scale_colour_manual(values = c("darkorange", "purple", "cyan4")) +
  labs(
    title = "Relationship Between Flipper Length and Body Mass",
    subtitle = "Palmer Penguins Dataset",
    x = "Flipper Length (mm)",
    y = "Body Mass (g)",
    colour = "Penguin Species"
  ) +
  theme_minimal() +
  theme(legend.position = "bottom")
p
```

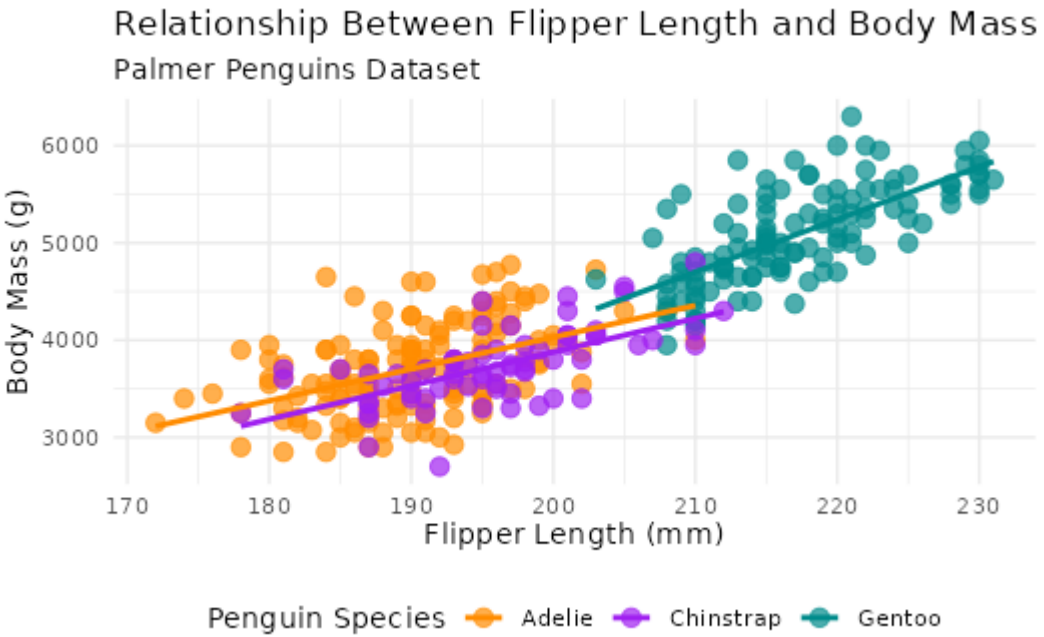

Relationship Between Flipper Length and Body Mass
Palmer Penguins Dataset
Body Mass (g)
6000
5000
4000
3000
170
180
190
200
210
220
230
Flipper Length (mm)
Penguin Species
Adelie
Chinstrap
Gentoo

Adding more layers

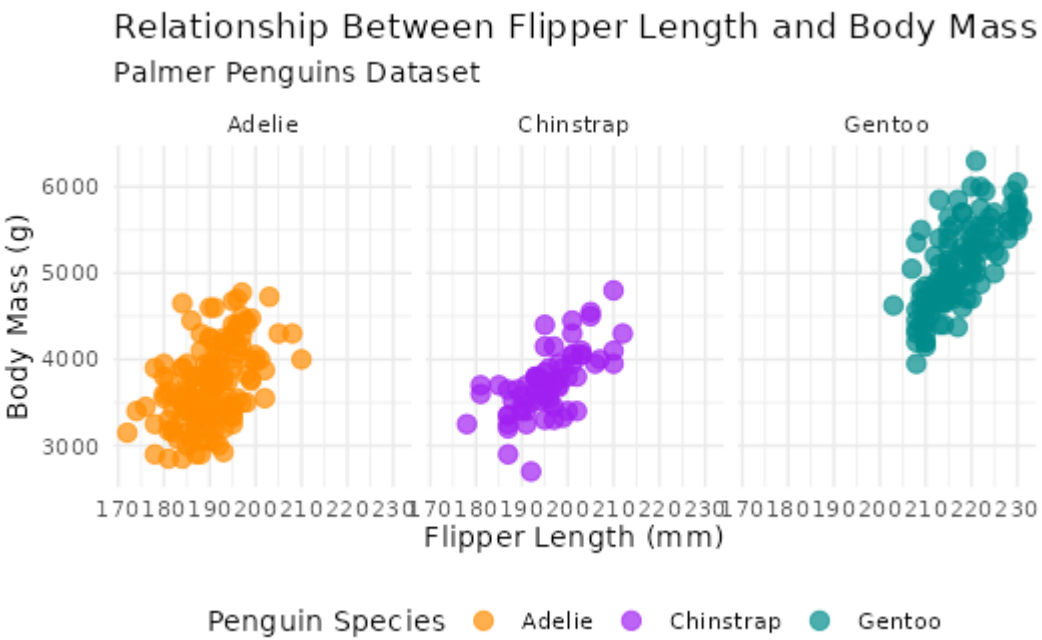
Once you have a base plot saved as `p`, you can keep adding layers. Trend lines and facets are two of the most useful:

```
# Add a linear trend line per species
p + geom_smooth(method = "lm", se = FALSE)
```

```
`geom_smooth()` using formula = 'y ~ x'
```




```r
# Split into separate panels
p + facet_wrap(~ species)
```


Relationship Between Flipper Length and Body Mass
Palmer Penguins Dataset
Adelie
Chinstrap
Gentoo
Body Mass (g)
6000
5000
4000
3000
170180190200210220230
Flipper Length (mm)
Penguin Species
Adelie
Chinstrap
Gentoo

References and resources

Further reading

- Wickham, H. (2016). *ggplot2: Elegant Graphics for Data Analysis*. Springer.
- Wickham, H., & Grolemund, G. (2017). *R for Data Science*. O'Reilly Media.
- ggplot2 documentation
- The R Graph Gallery

Thanks!

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